Solo Leveling Gates: Plan of Action

MDVLCraft, option C · 22 October 2026 · companion to the Solo Leveling Gates Design Report

What this plan builds. Item-summoned gates that lead into themed dungeon instances, in eight phases. Each phase ends in a playable build that is tested in game.

- **Keys:** crafted, one per rank, E to S.
- **Red gates:** a random roll on entry.
- **Dungeon breaks:** the mild version.
- **Rewards:** all three sources (mob drops, mana-crystal veins, boss chests).
- **Experience:** a clear bonus.
- **Late game:** challenges 1–4 (gate modifiers, red S gates, the Demon Castle tower, rule gates).

No hunter ranks. A player's strength is their gear and skill trees, and a gate's rank only describes the gate.

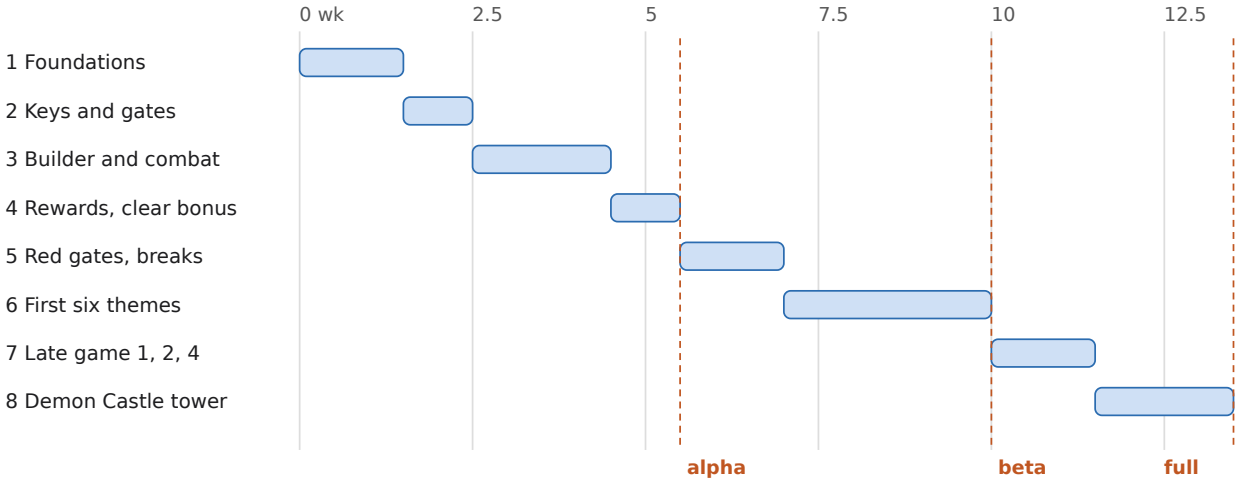
Estimated effort: about 13–14 developer-weeks, released as an alpha (after phase 4), a beta (after phase 6) and the full feature (after phase 8).

1. Design summary

Decision	Choice in this plan
World layout	Three dimensions, each cut into 1024×1024-block cells (one cell per open gate): • `mdvlcraft:gate_realm`: room dungeons. • `mdvlcraft:red_realm`: generated red-gate landscapes. • `mdvlcraft:demon_castle`: the tower. Themes are biomes laid out as a checkerboard, so every cell has its theme's sky, fog, particles and music without needing a dimension per theme (see 3.1).
Keys	Crafted Gate Keys per rank. A key can be attuned to a theme with a catalyst; otherwise the theme is random. The key is consumed when used.
Ranks	E, D, C, B, A, S map onto the quest book's six chapters. Nothing stops a player from using any key they can craft; the recipes are the gate.
Red gates	A random roll when the party first enters a B, A or S gate (5% / 8% / 12%, configurable). A red gate seals until its boss dies.
Dungeon breaks	Mild. A gate opened and not cleared within 2 real hours releases one wave of its mobs around itself, then closes.
Rewards	Essence stones from mobs, mana crystals from veins, and a per-player boss chest. Currency is spent at a Gate Forge.
Experience	Clear bonus to every party member present at the boss kill. Gate mob kills are exempt from the per-chunk anti-farm cap.
Late game	Modifiers on A/S keys, red S gates, an infinite Demon Castle tower with a boss every 5th floor and saved progress, and rule gates.

2. Roadmap

Eight phases, about 13.5 weeks; playable builds after phases 4, 6 and 8



Each phase has a done-when test in section 12. Phase 6 is mostly building rooms in game rather than code, so it can overlap phases 7 and 8 if someone else builds the rooms.

Phase	Ships	Weeks	Playable?
1 Foundations	Gate dimensions, cell allocator, teleport in/out, protection, cleanup	1.5	Debug only
2 Keys and gates	Gate Keys and recipes, the gate entity, ranks, party roster, opening and entering	1	Debug only
3 Builder and combat	Room planner, template placement, room-entry spawning, scaling, boss room and lock	2	One test theme
4 Rewards and clear bonus	Essence stones, crystal ore, boss chest, Gate Forge, clear experience	1	**Alpha**: Gravekeeper Crypt E–C
5 Red gates and breaks	Red roll, red realm landscapes, hazards, hidden boss, mild breaks	1.5	Yes
6 First six themes	About 70 room templates, 6 red landscapes, mob and boss pools, biomes and music	3	**Beta**: all ranks
7 Late game 1, 2, 4	Gate modifiers, red S gates, rule gates (2 rules at launch)	1.5	Yes
8 Demon Castle tower	Infinite floors, boss every 5th floor, saved progress, leaderboard	2	**Full**

3. Phase 1: Foundations

3.1 Dimensions (data files in the binder)

Dimension	Generator	Biomes	Dimension type

`gate_realm`	Noise generator with an empty ("air") noise setting, the same trick merlin's boss mods use, so the world is void	`checkerboard` source at scale 6 (64 chunks = 1024 blocks) cycling the theme biomes, so each 1024-block cell gets one theme	Fixed time (dusk), skylight on, ambient light 0.1, beds and respawn anchors off, y 0–256
`red_realm`	Real noise terrain (vanilla overworld-style settings, no structures, no ores, mob generation off)	The same checkerboard of theme biomes, whose surface rules make snow, sand, jungle floor and so on	Fixed time (deep dusk), beds off, y −64–320
`demon_castle`	Void, like `gate_realm`	One fixed biome: ash particles, red fog, castle music	Fixed time (night), beds off

Each theme biome sets sky colour, fog colour, water colour, ambient particles, background music and mood sounds. It has no spawners, because every mob comes from the gate. Checkerboard biomes are why three dimensions are enough instead of one per theme. The cost is that time of day is shared within a dimension.

Cell to theme rule. Minecraft's checkerboard picks a biome by `(cellX + cellZ) mod themeCount`. The allocator only hands a gate a cell whose index gives the gate's theme.

3.2 Cells and instances

- **Cells** are 1024×1024 blocks, aligned to region files (4 region files per cell). Cells start at a 2048-block offset from 0,0.
- **`GateInstances`** (SavedData on the overworld) stores each instance:
  - id, dimension, cell (x, z), theme and rank
  - state: `BUILDING` → `ACTIVE` → `CLEARED` → `CLOSED`
  - roster (UUIDs), entry point, the overworld gate position
  - room list with each room's bounds and markers, and the timers
- **Allocation:**
  - `gate_realm` cells are reused. On close, the recorded room bounds are filled with air a few rooms per tick, then the cell is marked free.
  - `red_realm` cells are never reused while the server runs, because generated terrain cannot be "cleared". The red realm is wiped on every server start instead.
- **Caps:** 2 open gates per player and 24 per server (config).

3.3 Moving players

- `GateTeleporter` (a Forge `ITeleporter` that returns a fixed arrival point and builds no portal) moves players in and out.
- On entry the player's origin is saved: dimension, position and facing. This reuses the save logic from `integration/PartyTravel`.
- Gate dimensions are **not** added to PartyTravel's follow-the-leader list, because gates manage their own party roster.
- Exit goes back to the saved origin, or to the gate's overworld position if the origin is gone.

3.4 Protection (`gate/GateProtection`)

- Block breaking and placing are cancelled in all three dimensions.
- Explosion block damage is cleared.

- Ender pearls, chorus fruit and enderman teleports are cancelled when they would leave the current room or landscape bounds.
- Teleports to other dimensions are cancelled while the player's instance is red and uncleared.
- A player outside their cell's bounds is pulled back to their last safe position. This covers glitching through walls.
- Waystones, homes and teleport spells: one compatibility check, cancelling `EntityTravelToDimensionEvent` and same-dimension teleports out of the cell. Pack mods with their own teleport items need to be checked one by one in testing.

3.5 Cleanup and safety

- **Server start:** delete the `red_realm` folder (the method Library of Exile uses), clear stale `BUILDING`/`ACTIVE` instances, and mark their gates as broken-closed.
- **Login:** a player inside a gate dimension whose instance is not `ACTIVE` is sent home.
- **Exclusions:** the three dimensions are left out of Chunky pre-generation and Distant Horizons. They are also listed in the server README so admins can skip them in backups.
- **Admin commands:** `/mdvl gate list | close <id> | tp <id> | open <rank> <theme> [red]`.

4. Phase 2: Keys and gates

4.1 Gate Keys (crafted)

The item is `mdvlcraft:gate_key` with NBT `rank`, optional `theme` and optional `modifiers`. Each rank has its own texture and glow colour. Recipes are shaped and shown in JEI. Ranks below E are not needed: the first key uses only overworld materials.

Key	Recipe (one key)	Chapter it unlocks in
E	4 iron ingots, 4 amethyst shards, 1 ender pearl	Wanderer
D	8 E essence stones, 4 E mana crystals, 1 gold block	Wanderer / Adventurer
C	8 D essence stones, 4 D mana crystals, 1 diamond	Adventurer
B	8 C essence stones, 4 C mana crystals, 2 blaze rods	Veteran
A	8 B essence stones, 4 B mana crystals, 2 netherite scrap	Champion
S	8 A essence stones, 4 A mana crystals, 1 netherite ingot, 1 nether star	Legend

Attunement. To pick a theme, craft the key with that theme's catalyst. Otherwise a random theme valid for the rank is chosen when the gate opens.

Theme	Catalyst
Gravekeeper Crypt	bone block
Frozen Hollow	packed ice
Goblin Warren	torch
Orc War Camp	iron axe
Mage Spire	lapis block
Infernal Forge	magma block

Keys can also come from boss chests (15% chance of the next rank up), which gives a shortcut without replacing crafting.

4.2 The gate

- `GateEntity` is placed where the key is used, after a check for 5×6 blocks of free space.
- **Size by rank:** E is 3 blocks tall and S is 9. It is drawn as a swirling ring (blue, or red once it rolls red) with particles and a hum, using the binder's existing custom renderer approach.
- **Stored on the gate:** instance id, rank, theme, owner and opening game time.
- **Tooltip on look:** rank, theme, time left before it breaks, and how many party members are inside.
- **Building:** opening the gate starts building the dungeon at once. The gate shows "stabilising" for about 5–10 seconds until the dungeon is ready, then players walk in.

4.3 Party roster

- **Who may enter:** the opener and anyone allied with them in Parties (`AllianceAPI`), up to the rank's party cap (E 2, D 3, C 3, B 4, A 5, S 6).
- **Normal gates:** members may leave and come back freely until the boss dies, as in the story.
- **Red gates:** the roster freezes the moment the gate turns red.

5. Phase 3: Dungeon builder and combat

5.1 Room templates

- **Size:** rooms are `.nbt` structure templates built in game. Each fits in a 48×48×48 slot, the structure block limit.
- **Room kinds per theme:** entrance, corridor (straight, turn, T), small combat, large combat, crystal room, treasure alcove and boss room (one per boss size class).
- **Doors:** each sits at the centre of a face, 3 wide and 4 tall, so any two rooms connect.
- **Markers:** data-mode structure blocks inside each template mark where things go.
  - `spawn:melee`, `spawn:ranged`, `spawn:elite`
  - `boss`, `crystal`, `chest`, `exit`, `door`
- **Recording:** the builder reads the markers and stores them in the instance, then removes the marker blocks.

5.2 Planner (`gate/DungeonPlanner`)

1. Pick a main path length from the rank: E 3–4 rooms, S 10–12.
2. Walk a grid of 48-block slots from the entrance and place rooms along the path. Corridors join them where turns are needed.
3. Add 0–2 side branches. One always holds the crystal room; the others can hold a treasure alcove.
4. End the path with the boss room.
5. The seed is the instance id, so a crash-rebuild gives the same layout.

Placement is spread over ticks: one room every 2 ticks, about 2–3 seconds for an S gate. This avoids lag spikes.

5.3 Spawning and scaling (`gate/GateSpawner`)

- **Trigger:** mobs spawn when a roster member first enters a room. Count = room markers × rank factor × (1 + 0.25 per extra party member).
- **Spawn method:** mobs are created the way `integration/BossArena` does it, by calling `finalizeSpawn` directly with no spawn event. This gets past the pack's content filter, which cancels event spawns for Legendary Monsters, Alex's Caves and the spell add-ons.
- **Tags:** each mob gets `mdvlcraft_gate` plus the instance id. It never despawns, and it is pulled back to its room if it leaves.
- **Scaling** uses attribute modifiers on spawn (health / damage / armour):
  - E ×1.0 / 1.0 / +0
  - D ×1.25 / 1.1 / +2
  - C ×1.5 / 1.2 / +4
  - B ×2 / 1.35 / +6
  - A ×2.5 / 1.5 / +8
  - S ×3 / 1.75 / +10
- **Health cap workaround:** vanilla caps max health at 1024. Above the cap, extra toughness becomes damage reduction applied in `LivingHurtEvent`.
- **Elites:** 1 per large room, with ×2 health, a glowing outline, a boss bar and guaranteed essence stones.

5.4 Boss room

1. When the party enters, the doors are swapped to sealed blocks, the way the Aether's boss rooms lock.
2. The boss spawns with gate scaling and a `ServerBossEvent` bar.
3. The boss's own loot is cleared in `LivingDropsEvent`; the gate's reward table replaces it in phase 4.
4. On death the doors reopen, an exit portal appears at the boss marker, and the instance becomes `CLEARED`.
5. The gate then stays open for 2 minutes and collapses, sending anyone left inside home.

Bosses are copies of pack bosses, so quest-book lairs are unaffected. Each boss is listed with a size class (small, medium or large), which picks the boss-room template that fits it.

5.5 Death and logout

- **Death in a normal gate:** the player respawns outside and may walk back in.
- **Death in a red gate:** the player respawns outside and cannot return.
- **Whole party dead:** the gate fails and closes, and the key is lost.
- **Logout:** the player stays on the roster. If the gate closed while they were gone, the login check sends them home.

6. Phase 4: Rewards and clear bonus

Source	How it works

Essence stones (mobs)	Seven items, one per rank: E grey, D white, C green, B blue, A red, S purple, plus black for hidden bosses. They drop through a global loot modifier with a new condition, `mdvlcraft:gate_mob` (matches the tag and reads the instance rank). Drop rates: normal mob 35% for 1, elite 100% for 2–3, boss 8–12. Break mobs drop none. Listed for JER through `loot/CodeDrops`.
Mana crystals (veins)	Six ore blocks, one per rank, placed at crystal markers: 12–20 per vein, more in red gates. Each needs a pickaxe tier by rank: E–D stone, C iron, B diamond, A–S netherite. Fortune works. They drop mana crystal items. Mining is optional and takes time, like the story's mining teams.
Boss chest	A `GateRewardChest` block at the boss marker. Each roster member present at the kill gets their own roll from `loot_tables/gates/<rank>/boss.json`. The chest stays usable for the 2-minute collapse, and leftovers go to the player's inventory or drop at their feet. Contents: essence, crystals, enchanted books, trophies, a 15% chance at a next-rank key, rank gear materials and the Return Stone.

6.1 Gate Forge

The Gate Forge is a crafting block (iron + amethyst + one A essence stone) with its own recipe type. It adds no new armour models; it spends currency on four things:

- **Gear tempering:** adds a gate bonus to an existing weapon or armour piece, from E to S (+2% to +12% damage or damage reduction against gate mobs, plus a small general bonus at A–S), stored in NBT.
- **Return Stone:** escapes a normal gate instantly. It does not work in red gates or the tower.
- **Modifier reroll** on A/S keys (phase 7).
- **Tower sigil** (phase 8).

6.2 Clear bonus experience

- **Clear bonus:** on boss death, each roster member inside the instance gets skill-tree experience through the puffish skills API.
  - E 200, D 400, C 800, B 1,500, A 3,000, S 6,000.
  - Red ×1.5; with modifiers, +15% per modifier.
- **Kill experience:**
  - The normal kill source in `vocation_parts/experience.json` gets a condition that excludes gate mobs.
  - The binder gives gate kills their usual amount (dropped XP + max health / 5, or 150 for mages) directly.
  - This skips the 20-kills-per-chunk cap, which dense rooms would otherwise hit within seconds.
- **Sharing:** kill experience is shared through the existing `content/PartyExperience`. The clear bonus is not shared, because everyone present already gets it in full.

7. Phase 5: Red gates and mild dungeon breaks

7.1 The red roll

1. The first time a roster member steps into a B, A or S gate, roll the red chance (B 5%, A 8%, S 12%; modifiers can raise it).
2. On red: the gate turns red, the room dungeon is discarded (its cell is freed), and a `red_realm` cell of the same theme is used instead.

3. The party is pulled in together. The roster freezes and the gate refuses anyone new.

7.2 Red landscapes

- **The land:** the cell's terrain is ordinary generated land in the theme's biome (snowfield, desert, jungle and so on). A 96-block-radius area around the arrival point is playable, walled by an invisible boundary that pulls players back.
- **The boss lair:** a hand-built template per theme, pasted 60–90 blocks from arrival in a random direction.
- **Packs and camps:** 6–10 scattered small templates (camps, ruins, nests) with spawn markers.
- **Mobs:** one rank above the key.
- **Hazard** per theme, applied every 5 seconds:
  - Cold: Slowness I, and freezing damage without leather or fur-lined armour or a campfire nearby.
  - Heat: hunger drain ×2.
  - Spores: Poison I for 3 s when standing in jungle grass.
- **Food:** hunger runs faster, and food is scarce apart from what mobs drop, echoing the story's starving snowfield party.
- **Hidden boss:** after the lair boss dies, 50% chance (100% for S) that a second boss rises. It drops black essence stones.
- **Exit:** nobody leaves until the last boss dies. Then the exit portal opens at the lair and the 2-minute collapse starts.

7.3 Mild dungeon breaks

- **Timer:** 2 real hours from opening (decided), shown on the gate and changeable in config. It is measured in overworld game time, so it keeps running while the gate's chunk is unloaded. If the chunk is unloaded at break time, the break happens the next time it loads.
- **Warning:** at 10 minutes left, the gate pulses and players within 128 blocks and the owner get a chat warning.
- **The break:** the gate spawns one wave around itself: 6 + 2 per rank mobs from its pool, at its rank scaling, tagged as break mobs (no essence, no skill experience). Then the gate closes and the key is lost.
- **Gates nobody has entered** break the same way.
- **Server option:** breaks can be turned off in the binder config.

8. Phase 6: The first six themes

Theme	Ranks	Bosses by rank (copies)	Content to build
Gravekeeper Crypt	E–A	Aptrgangr (C), Dead King (B), Enraged Dead King (A)	12 rooms, 1 red landscape (cursed marsh)
Goblin Warren	E–C	Kobolediator (E–D), twin Kobolediators (C)	11 rooms
Frozen Hollow	D–A	Frostbitten Golem (D–C), Maledictus (A)	11 rooms, 1 red landscape (snowfield, ice-elf camps)
Orc War Camp	C–A	Lupus Rex alpha (C), Ignited Revenant (B), Possessed Paladin (A)	11 rooms, 1 red landscape (badlands)

Mage Spire	D–A	Ancient Knight (D–C), Jade Executioner (B), Spell Warder (A)	12 rooms, 1 red landscape (ruined city)
Infernal Forge	B–S	Lava Eater (B), Ignis or Netherite Monstrosity (A), Echo of Tyros (S)	12 rooms, 1 red landscape (lava fields)

S-rank gates at launch use the Crypt (Enraged Dead King plus a hidden boss) and the Infernal Forge. Later themes reuse the same pipeline: Sunken Naga Temple, Desert Pyramid, Primeval Jungle, Insect Nest and Void Rift.

Each theme is one data file, `data/mdvlcraft/gate_themes/<theme>.json`, which lists:

- biome and room template pools
- mob pools per rank band, with weights
- bosses per rank, with size class
- hazard, catalyst item, red landscape and lair templates
- music

Adding a theme needs no code.

9. Phase 7: Late-game challenges 1, 2 and 4

9.1 Gate modifiers (challenge 1)

- **Rolling:** A keys roll 0–1 modifiers and S keys roll 1–3, at crafting time. They are shown in the key's tooltip and can be rerolled at the Gate Forge for A essence.
- **Data:** each modifier is a data entry, `data/mdvlcraft/gate_modifiers/*.json`.
- **Reward:** each modifier adds +20% essence and crystals and +15% clear experience.

Modifier	Effect
Enraged	Mobs deal +30% damage
Swarming	+50% mobs per room
Cursed	No natural regeneration inside
Volatile	Mobs burst on death (harmless to blocks)
Silenced	Spells cost double mana
Elite Pack	Every combat room has an elite
Crimson	Red chance ×2
Ironclad	Mobs +10 armour

9.2 Red S gates (challenge 2)

Red S gates come from the normal 12% roll, raised by the Crimson modifier. They always have a hidden boss, double crystal veins and two lair camps. They are the only source of black essence stones, which are needed for S tempering and the tower sigil.

9.3 Rule gates (challenge 4)

- **When:** an S gate has a 6% chance to become a trial gate, after the red roll fails. The rule is shown as a title on entry and on a stone tablet in the entrance.
- **Rewards:** trial gates give ×1.5 rewards and a trophy.

- **At launch:** two rules, cheap to build.
  - *No Potions:* drinking potions is cancelled.
  - *No Healing:* healing from any source except food is cancelled.
- **Later:** two richer rules, each its own mini-project.
  - *Statues:* a statue entity that moves only while no player looks at it. This needs a new entity.
  - *The Narrowing Door:* a door that shrinks each time a member leaves, so someone must stay behind.

10. Phase 8: The Demon Castle tower (challenge 3)

Summary. The tower has no top. Floors are built only when someone reaches them, so it never takes up more than two floors of space per run. Every 5th floor is a boss floor. Each player's progress is saved, so they return to the last floor they reached instead of starting again at floor 1.

10.1 Getting in

- **Castle Sigil:** crafted at the Gate Forge from 4 black essence stones, 8 S essence stones and 1 netherite ingot. It is not consumed: using it opens a dark-red castle gate in front of the player, the same way a key opens a gate.
- **Party:** the opener's Parties allies may join, up to 6, as with gates.
- **Instance:** the run gets a cell in `demon_castle`. Tower cells are reused like `gate_realm` cells.

10.2 Infinite floors without infinite space

Only the current floor and the next one exist; the previous floor is cleared behind the party



- Each floor is a small dungeon (landing, 2–4 rooms, stair room) at its own height band in the run's cell.
- Floors 5, 10, 15 ... are a single boss arena; the stairs only open when the boss dies.
- Layout comes from hash(world seed, floor number): floor 37 always looks the same, its mobs are fresh.
- Floors stack in two height bands (y 20–120 and 130–230) that alternate, so n and n + 1 never overlap.

- **A normal floor** is a landing, 2–4 rooms and a stair room. All rooms on a floor come from one theme; the theme changes every 5 floors and cycles through the six themes in order.
- **A boss floor** (every 5th) is one large arena with a boss from the tower boss list:
  - floors 5–20: Champion-chapter bosses
  - floors 25–50: Legend-chapter bosses
  - floor 55 and up: Legend bosses with +2 modifiers and a 25% chance of a hidden boss
- **Stairs** lead up. Walking onto the top step teleports the party to the next floor's landing. The next floor was built while they fought the current one, so the move is instant.
- **Scaling** starts at A rank. Each floor adds +5% health and +4% damage, compounding gently: floor 50 is about ×11 health, so past the 1024 health cap it switches to damage reduction (see 5.3).
- **Extra modifiers:** every 10 floors adds one permanent tower modifier (from the phase 7 list) for that stretch, shown on the landing tablet.
- **Space:** because only two floors exist at once, a run's cell never fills, however high the party climbs.

10.3 Saved progress: returning to your last floor

Piece	Design
`TowerProgress` (SavedData on the overworld)	Per player UUID: `reachedFloor` (the highest floor they have stood on), `checkpoint` (the floor after the last boss floor they cleared: 1, 6, 11 ...) and `bestFloor` (record, for the leaderboard). Saved to disk immediately on change, so a crash loses nothing.
When progress is written	Stepping onto a floor's landing sets `reachedFloor` to that floor if higher. Killing a boss on floor 5k sets `checkpoint` to 5k + 1.
Where you start	Using the Castle Sigil opens a menu: *Continue from floor N* (N = `reachedFloor`) or *Start from checkpoint* (to farm lower floors). The chosen floor is built fresh: its layout comes from the floor seed and its mobs are new, so leaving mid-floor and returning means fighting that floor from its landing again.
Leaving on purpose	A Leave Tower lever on every landing and in every stair room sends the player home. Progress is kept at `reachedFloor`, the last floor they were on.
Dying	**Drop back (decided).** A player who dies in the tower has `reachedFloor` set back to their `checkpoint`, the first floor of the current block of five. Dying on floor 23 returns them to floor 21 next time; dying on floor 25, the boss floor, returns them to floor 21 as well, because floor 25's boss was not beaten. The player is sent home and the rest of the party carries on. Their `bestFloor` record is never lowered.
Parties with different progress	The party starts at the **lowest** `reachedFloor` among its members, so nobody skips floors they have not earned. Everyone's progress rises as the party climbs. A strong player helping a friend therefore re-climbs from the friend's floor but loses nothing of their own, because their progress only goes up unless they die.
Logout or crash mid-floor	The run continues for the rest of the party. On login the player is sent home (the login check from 3.5). Their `reachedFloor` is already saved, so they resume from that floor next time.
Admin tools	`/mdvl tower progress <player> [set <floor>]`, `/mdvl tower top` (top 10 by `bestFloor`), `/mdvl tower close <run>`.

10.4 Tower rewards

- **Normal floors:** essence and crystals at A–S rates. A small chest in the stair room gives every member their own roll (the per-player chest from 6).
- **Boss floors:** a full boss chest plus Tower Seals, a tower-only token spent at the Gate Forge on the top tempering tier and cosmetic trophies.
- **First clear of a boss floor** (by that player's `bestFloor`): double seals and a clear bonus of 6,000 skill experience × (1 + floor / 50).
- **Anti-farm:** floors below a player's `bestFloor` − 10 give half rewards.
- **Leaderboard:** highest floor reached, shown with `/mdvl tower top` and as a line on the status page.

11. Code layout (binder)

Package / file	Contents

`gate/GateRank`, `GateTheme`, `GateModifier`	Rank table (scaling, rooms, party cap, red chance); theme and modifier data loaders (JSON reload listeners)
`gate/GateKeyItem`, `GateEntity`, `GateRenderer`	Keys, the gate in the world, its look
`gate/GateInstances`, `CellAllocator`, `GateTeleporter`	Instance SavedData, cells, moving players
`gate/DungeonPlanner`, `DungeonBuilder`, `RedLandscape`	Layout, tick-spread template placement, red landscapes and lairs
`gate/GateSpawner`, `GateBossRoom`, `GateHazards`	Room-entry spawning, scaling, boss lock and bar, red hazards
`gate/GateProtection`, `GateCleanup`, `GateBreaks`	Events, startup wipe, login check, break timers
`gate/GateRewards`, `GateForge*`, `GateRewardChest`	Essence and crystal items and ores, loot condition, per-player chest, forge block and recipes, clear experience
`gate/tower/TowerRun`, `TowerProgress`, `FloorPlanner`	Tower runs, saved progress, floor generation
Data	`dimension/`, `dimension_type/`, `worldgen/biome/gate_*`, `structures/gates/<theme>/*.nbt`, `gate_themes/`, `gate_modifiers/`, `loot_tables/gates/`, recipes, lang, textures, sounds

Network: one new packet for the gate tooltip and timer (the gate entity's synced data covers the rest), so the protocol number rises by one.

12. Testing at each phase

Phase	Done when (checked in game on the test server)
1	`/mdvl gate open` creates a cell; a player teleports in and out. Breaking, placing, pearls and Waystones are all refused. A restart wipes the red realm, and a logged-out player comes back home.
2	Every key crafts and shows in JEI. A key opens a gate of the right size and rank. Allies can enter; non-allies and an over-cap party member cannot.
3	A test theme builds 4–12 rooms with no lag spike over 50 ms. Mobs spawn on entry, including filtered mods' mobs. Scaling is right. The boss room locks and unlocks, and the exit portal and collapse work.
4	Essence drops at the right rates. Ore needs the right tier. Each member gets their own boss chest roll. The forge tempers gear. Clear experience reaches every member present, and gate kills are not capped.
5	A forced red roll (`/mdvl gate open B crypt red`) traps the party until the boss dies. The hazard ticks and the hidden boss appears. A gate with its timer shortened to 1 minute breaks into one wave and closes.
6	Every boss of every theme spawned and killed in its room at its ranks. Every room checked for exits and spawn markers.
7	Each modifier visibly applies. Both rules are enforced. A red S gate drops black essence.

8	Climbing floors 1 to 11 runs smoothly. A boss on floor 5 opens the stairs. Leaving on floor 7 and re-entering offers floor 7. Dying on floor 23 sends the player home, and they re-enter at floor 21. A two-player party with progress 3 and 12 starts at floor 3, and both rise. A restart mid-run keeps progress. `/mdvl tower progress set 100` builds floor 100 at the right scaling.

13. Risks and open questions

Risk	Plan
Bosses that need room or break blocks (Cataclysm, Legendary Monsters)	Boss size classes and arena templates sized per boss; block-damage events are already cancelled. Any boss that misbehaves in testing is dropped from the pool.
Many parties at once	Caps of 2 per player and 24 per server; room-entry spawning; tick-spread building.
World size	Gate realm cells reused, red realm wiped on start, tower keeps only two floors.
Escapes through other mods' teleports	Generic event cancels plus a per-mod check in phase 1 testing.
Balance against quest-book progression	Rewards are their own currency and tempering, not more chest loot. Tune after the alpha.
Room building effort	About 70 templates for six themes is the bulk of phase 6. It can be shared with a builder.

Decisions made (22 October 2026):

- **Tower death:** drop back to the start of the current block of five floors (10.3).
- **Break timer:** 2 real hours (7.3).
- **Trading keys:** keys can be traded between players (the default; no objection raised).
- **Quest-book compasses:** fixed separately, ahead of the gates. The quest book no longer gives compasses to Cataclysm lairs. It gives the lair's Cataclysm eye (Mech, Monstrous, Flame, Desert, Cursed, Abyss, Void or Storm Eye) instead, which, sneak-used, takes the party into that lair's own world. The boss quests now explain this.